The late losses of the mixture fit: theory, predictions and the experiments that decide

Version-4 basket model, branch `v4-mixture`

29 October 2026

The mixture form (`docs/mixture_theory.pdf`) did what it was introduced for: one posterior peak per trip, exact generation, a reliable likelihood estimate, and a better held-out score than the old form. This note is about the *training*. After its best step the fit loses held-out likelihood, on synthetic and on real data, and its generated baskets drift in size. Section 1 fixes notation. Section 2 lists what has been measured. Sections 3–6 take each mechanism that could cause the losses, say what it predicts, and say which measurement already decides it. Section 7 is the experiment plan, with the result each mechanism predicts stated before any run.

1 Setting and notation

Data. Trips t, each with a basket S_t and a household $h(t)$. The fit uses T training trips: $T = 7{,}695$ in the synthetic world `disc_c48_s3` (928 households, about 8 trips each) and $T = 158{,}983$ on Dunnhumby (1,920 households, about 83 trips each). 1,024 further trips are held out.

The model. A trip draws a taste $z \in \mathbb{R}^r$ ($r = 16$) from $\mathcal{N}(0, I)$, then a basket from the exact law given z:

$$p(S) = \int \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz, \qquad p(S \mid z) = \frac{e^{E_z(S)}}{Z_z}. \tag{1}$$

For a basket S with n_c products in category c and k categories present,

$$E_z(S) = \sum_{j \in S} \Big(b_j + \lambda_j - \tfrac{1}{2}|\varphi_j|^2 + z \cdot \varphi_j \Big) - \sum_c (n_c - 1)_+ \Big(z \cdot \psi_c - \tfrac{1}{2}|\psi_c|^2 \Big) - \sum_c \big[\rho_c \tbinom{n_c}{2} + f_c(n_c) \big] - \rho_0(k). \tag{2}$$

The symbols: b_j is product j's utility from the household, context and price terms; λ_j is its intercept; $\varphi_j \in \mathbb{R}^r$ is its taste loading; $\psi_c \in \mathbb{R}^r$ is the loading charged for each *extra* product of category c; ρ_c, f_c penalise several products of one category; $\rho_0(k)$ is the size term, indexed by the number of categories k. Z_z normalises over all baskets.

Parameter blocks. *Household*: θ_h, γ_h (40 numbers per household). *Product context*: the per-product context and price factors ($\alpha, \beta, \mu, \zeta, \dots$). *Interaction*: φ, ψ, λ. *Size*: ρ_0, ρ_c, f.

The training procedure (`scripts/fit_mixture.py`). Each step takes a batch of 256 training trips and does the following.

1. It finds each trip's posterior peak, draws $K = 8$ tastes $z_1, \dots, z_K$ from a proposal q, and forms the importance bound

$$L_K(S) = \log \frac{1}{K} \sum_{k=1}^{K} w_k, \qquad w_k = \frac{\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)}{q(z_k)}. \tag{3}$$

   The proposal is $q = 0.8\,\mathcal{N}(m, 1.2^2\Sigma) + 0.2\,\mathcal{N}(m, I)$, where m is the peak and Σ the Laplace covariance. It is deliberately wider than the posterior.

2. It takes one Adam step on $-\overline{L_K}$ + prior$/T$ for every block.

Every 30 steps it also does the following.

3. *Moment matching*: an extra step $\lambda_j \mathrel{-}= 0.35\,\bar{g}_j/r_j$, and the same for ρ_0, ρ_c with factor 0.5. Here $\bar{g}$ is the gradient averaged over the 30 steps and r the data rate.

4. *Whitening*: it estimates the pooled posterior second moment $M = \frac{1}{B}\sum_t \mathbb{E}[zz^\top \mid S_t]$ on 256 fixed trips ($K = 256$). With $A = M^{1/2}$ it sets $\varphi_j \leftarrow A\varphi_j$, $\psi_c \leftarrow A\psi_c$ and $\lambda_j \mathrel{+}= \frac{1}{2}(|A\varphi_j|^2 - |\varphi_j|^2)$. Then it clears Adam's state for φ, ψ.

Held-out score means the mean of $\log p(S)$ over 1,024 held-out trips with $K = 256$. *Training score* means the same over 1,024 training trips. All comparisons are paired, trip by trip, with the same random numbers.

2 What has been measured

	synthetic (step 420 → 1,210)	real (step 360/240 → 1,244)
held-out score	-0.18 ± 0.04	-0.66 ± 0.08 (from step 240)
training score	$+1.31 \pm 0.04$	-0.56 ± 0.06 (from step 240)
generated size (data)	8.68 → 7.92 (8.69)	8.0–8.3 → 7.5–7.9 (7.26)
$\frac{1}{2}\lvert\varphi_j\rvert^2$, mean over products	0.59 → 1.75	0.65 → 2.33
growth of $\lvert\varphi_j\rvert^2$ by purchase count	×2.2–3.7 in every band	×1.6–2.9 in every band
λ mean change	+0.65	+1.02
$\rho_0(k)$ change	up to +4.4	up to +15.3
product of whitening trace$/r$	×3.0 (40 whitenings)	×1.54 (29 whitenings)

Further measurements:

1. *Block swaps, synthetic.* Putting the household block back to step 242 gains +0.54 on held-out trips and loses 0.61 on training trips. The product-context block gains +0.30 and loses 0.69. Both together, back to step 420, reach −26.43.

2. *One whitening step* applied to the real step-360 fit changes the score by -0.007 ± 0.005 (held-out) and -0.010 ± 0.005 (training). The same step with $1.5M$ costs −1.3 and doubles the generated size.

3. *The estimate of M* is unbiased at $K = 256$: trace$/r$ is 1.020 at $K = 256$ and 1.019 at $K = 2{,}048$ (synthetic step 420). At $K = 16$ it is 1.037.

4. *The pooled posterior, real:* its mean has norm 0.21–0.23, its covariance eigenvalues lie in 0.80–1.16, and its trace$/r$ is 1.00.

5. *The bound gap* $\log p - L_8$ is the same for every basket size, at 0.14 nats at step 1,210.

3 Mechanism 1: memorisation by per-household and per-product parameters

Theory (`docs/overfitting_theory.pdf`). A block with d free numbers per household, fitted to that household's n_h trips, makes the training score better than the held-out score by about $d/(2n_h)$ per trip. With $d = 40$ and $n_h \approx 8$ that is about 2.5 nats. Early stopping stops this partway.

Why the hierarchical prior failed. `v4/hier_prior.py` centres each block column at its mean over rows. The base model had already fitted every row with early stopping on validation. A prior at the column mean pulls those validated row values toward the average, so it removes real signal (measured: −0.3 from the first step on). For γ and β, a single household's trips barely determine the values, so the EM update drives their prior scale to zero.

The fix that follows. The row values to keep are the base model's, $x^{(0)}$. Use the penalty

$$\frac{\tau}{2}\sum_{r,k}\bigl(x_{rk}-x^{(0)}_{rk}\bigr)^2 \quad \text{(in total log-likelihood units).} \tag{4}$$

For a row whose own likelihood has curvature a along a coordinate, the penalised fit moves that coordinate by the fraction $a/(a+\tau)$ of the unpenalised move. This is a smooth version of early stopping: $\tau = 0$ is the present fit and $\tau = \infty$ freezes the block. τ is chosen on validation. It is not a model of the population: $x^{(0)}$ was itself fitted to the training trips. It only limits how far the mixture fit moves these blocks.

Predictions. Synthetic: freezing or anchoring the household and product-context blocks removes most of the held-out loss after step 420; the best held-out score is at least the swap result -26.43. Real: this mechanism cannot explain the real loss, because there the training score fell as well. Memorisation raises the training score.

4 Mechanism 2: the steps that are not gradient steps

The real fit loses on held-out *and* training trips. Steps 1–2 alone (stochastic gradient ascent on $\overline{L_K}$) can lose training likelihood only through noise or through the difference between L_K and $\log p$ (Section 5). Steps 3–4 are not gradient steps.

Whitening (parameter-expanded EM). Write $z = Az'$ with $z' \sim \mathcal{N}(0, I)$. Then $z \cdot \varphi_j = z' \cdot A\varphi_j$, and λ absorbs $-\frac{1}{2}|\varphi_j|^2$ exactly. So the new model with prior $\mathcal{N}(0, I)$ equals the old model with prior $\mathcal{N}(0, M)$, except for one term. In (2), the term $+\frac{1}{2}(n_c-1)_+|\psi_c|^2$ becomes $+\frac{1}{2}(n_c-1)_+|A\psi_c|^2$. It is linear in the number of extra products, and f_c (free only for $n_c = 2, 3, \geq 4$) and ρ_c (quadratic) cannot absorb it. A second omission: the EM update of a Gaussian prior also moves its mean, and $A = M^{1/2}$ folds the pooled mean $\bar{\mu}$ into the scale. *Measured:* at the real step-360 fit, one step costs 0.007–0.010, and $|\bar{\mu}| = 0.21$ shifts each product's utility by an amount with standard deviation 0.11. So whitening alone, at 29 steps, can account for at most about 0.3 of the 0.66, and only if nothing is recovered in between.

The reset of Adam's state. After a reset, Adam's first step moves every coordinate of φ, ψ by $\pm$ the learning rate, whatever the size of its gradient. With resets every 30 steps, this adds a random walk of about $0.003\sqrt{29} \approx 0.016$ per coordinate. That is far below the measured change ($|\Delta\varphi_j|^2 \approx 1.3$ after the best rotation). *Decided: not the cause.*

Moment matching. For λ_j, the gradient of the mean log-likelihood is $r_j - \hat{r}_j$: the data rate minus the model rate, averaged over the posterior. Dividing by r_j gives a damped Newton step, and its fixed point is the same as the gradient's. There are two departures:

1. it uses the gradient averaged over the last 30 steps, while Adam keeps moving the same parameters, so the two can overshoot together along directions where λ, ρ_0 and the scale of φ trade off;

2. its noise. A product bought c times in 7,680 trips gets a relative error of $1/\sqrt{c}$, so λ_j jitters with standard deviation about $0.35/\sqrt{c}$ around its target.

The size terms have the same structure. The measured change is a coordinated drift: $|\varphi|$ grows in every frequency band, λ rises by about 1, and ρ_0 rises by up to 15. That is the trade-off direction of the first departure.

Predictions. If moment matching drives the loss, a continuation without it loses less on training and held-out trips. If whitening drives it, a continuation without whitening does. Both continuations still train every parameter by the gradient, so they are fair tests.

5 Mechanism 3: training on the bound instead of the likelihood

The training gradient is

$$\nabla L_K = \sum_k \tilde{w}_k \nabla \log\big[\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)\big], \qquad \tilde{w}_k = w_k \Big/ \sum_l w_l,$$

because the proposal q does not depend on the parameters. This is the self-normalised estimate of $\nabla \log p(S) = \mathbb{E}[\nabla \log p(S, z) \mid S]$. Its bias is of order $1/K$, and it points toward averages under q. Since q is wider than the posterior, the gradient behaves as if each posterior were wider than it is. *Measured:* at $K = 16$ the pooled second moment is overstated by 1.8% (1.037 against 1.019); at $K = 8$ it is about twice that. The fixed point of L_8 is therefore not the maximum-likelihood point, and the gap grows as posteriors narrow (synthetic: 0.06 → 0.14 nats).

Prediction. A continuation with $K = 32$ (bias four times smaller) loses less. This costs about 4 times as much per step, so it is the last experiment, run only if Sections 3–4 leave a loss unexplained.

6 Mechanism 4: generated size

At a stationary point of the likelihood, the size statistics match the data *averaged over each trip's posterior*:

$$\frac{1}{T} \sum_t \mathbb{E}\big[\, \mathbb{E}[n \mid z] \;\big|\; S_t \big] = \bar{n}_{\text{data}}.$$

Generation draws $z \sim \mathcal{N}(0, I)$. The two agree when the pooled posterior $\frac{1}{T}\sum_t p(z \mid S_t)$ equals $\mathcal{N}(0, I)$. That holds at the maximum-likelihood point of a *correctly specified* model, with large T. Whitening forces only the second moment to I. *Measured* (real): the mean is not zero ($|\bar{\mu}| = 0.22$), and the covariance has eigenvalues from 0.80 to 1.16. So the size gap is a symptom, not a separate cause. It shrinks if Sections 3–5 bring the fit closer to the maximum-likelihood point. If a gap remains, it measures misspecification.

7 Experiments

All runs use the existing code with three new switches: `--no-moment-match`, `--no-whiten`, and `--anchor BLOCKS --anchor-prec` τ (penalty (4); $\tau = \infty$ is written `--freeze`). Scores are paired, on 1,024 held-out and 1,024 training trips with $K = 256$. Selection is on held-out trips only; the test weeks are never opened.

Real (from the step-360 checkpoint, 480 steps to step 840, same batches).

run	change	prediction if its mechanism is the cause
R0	none (control)	held-out and training both below step 360
R1	`--no-moment-match`	loss much smaller than R0 on both splits
R2	`--no-whiten`	loss much smaller than R0 on both splits

Synthetic (fresh 10-epoch fits, the same seed as the original run, which is the control).

run	change	prediction
S1	freeze household and product context	best held-out ≥ -26.43; small late loss
S2	anchor, $\tau = 10$	between S1 and the control
S3	anchor, $\tau = 100$	close to S1
S4	`--no-moment-match`	tells whether Mechanism 2 also acts here
S5	`--no-whiten`	likewise

The value $\tau = 10$ is about the measured likelihood curvature of a household coordinate with 2–4 trips (6–16), so it halves the movement of such a household.

Reading the results. For each run, report the held-out and training score at the best and final steps, against the control. Also report the generated size and the change in ρ_0 and $|\varphi|$. A mechanism is confirmed only if switching it off changes the late loss by more than twice the paired standard error. Otherwise it is rejected.